

FlowState

A strategy is real and robust.
How much money can it take?

Ashwin Jain

Department of Computer Science and Engineering
Motilal Nehru National Institute of Technology Allahabad
Prayagraj, India
ashwinaadi2007@gmail.com

A frozen benchmark for deployment capacity and alpha decay, measured under the information that is actually observable — and a principal negative result on what is not.

Code, corpus, fixtures and evaluation protocol:

github.com/Ashwin-aadi/Trivijaya-Quant-Research

Programme overview and results: ashwin-aadi.github.io/Trivijaya-Quant-Research

8 August 2026

Abstract

The first question a portfolio manager asks about a strategy is how much money it can take. The standard answer is a capacity curve, and building one requires knowing how far a given order moves the price. **We measure whether daily bars can supply that knowledge, and report that within the estimators and information available from daily OHLCV the obstruction is one of identification rather than statistical precision.** Two equally defensible weighting schemes applied to the same reversal regression disagree about the *sign* of transient impact at 3 of 4 horizons, and nothing in the data adjudicates between them. Amihud illiquidity, computed on the same panel as a control, is stable across a sample split at 0.836 while the fitted impact exponent reaches only 0.374: the failure is in the model, not in the data.

We therefore release FlowState, a frozen benchmark that reports what daily bars do support. Deployment capacity is redefined as a *constraint*—the largest AUM at which a strategy can place every one of its trades inside a stated participation limit—which is arithmetic on observed turnover and assumes no impact coefficient. On a survivorship-free Indian liquidity universe of 184 symbols over 2020–2024, standard price-and-volume factors deploy Rs.0.81–7.91 crore, and in every case the binding constraint is the least liquid position the strategy holds rather than the average liquidity of its book.

Two further results are null. **No factor's edge decays measurably** across horizons from 1 to 63 sessions, and none is statistically established at the shortest horizon, so the precondition for a decay study fails. **Deployable size does not collapse when foreign participation reverses**—the effect this benchmark was designed to detect—with outflow-to-inflow ratios between 0.94 and 1.24. That null rests on a derivatives-based flow proxy and cannot distinguish a stable capacity from an uninformative proxy; we state this in the results rather than the discussion.

Finally, a result about benchmarks rather than about markets. Applying the finished benchmark to 156 machine-generated strategies *before* publishing it exposed two defects that 5 standard factor strategies had not, and repairing them moved every capacity figure *downward* by between a factor of four and a factor of twenty-six.

Keywords: deployment capacity • market impact • alpha decay • participation constraints • negative results • Indian equities

Key findings

Each number answers a question, carries its sample size, and is tagged with the population it describes.

1. How much can a standard factor actually deploy?

[GENERATOR-INDEPENDENT]

Rs. 0.81–7.91 crore, across 5 factors on a survivorship-free universe of 184 symbols. We are not aware of these figures being openly published for Indian equities. They are *constraint* figures — the largest book that can place every trade inside the participation limit — and never impact-erosion figures.

2. What sets that number?

[GENERATOR-INDEPENDENT]

The single least liquid position, not the average liquidity of the book. Capacity computed from the median session overstates the binding figure by $4.5\times$ to $26.2\times$. **The weakest position determines the capacity of the entire portfolio**, and for one factor a single name accounts for 7.7% of the constraint.

3. Can daily data price market impact?**[GENERATOR-INDEPENDENT]**

No, and this is a principal finding rather than a limitation. Two equally defensible weightings of the same regression disagree on the *sign* of impact at 3 of 4 horizons. The order-size region a capacity study asks about contains 2 of 208,042 observations. The obstruction is identification, not precision.

4. Is that just noisy data?**[GENERATOR-INDEPENDENT]**

No — the control on the same bars is stable. Amihud illiquidity, computed on the same panel and split the same way, reaches 0.836 while the fitted impact exponent reaches 0.374 ($n = 166$). **The failure is in the model, not in the data.**

5. Does capacity collapse when foreign money leaves?**[GENERATOR-INDEPENDENT]**

No. Outflow-to-inflow capacity ratios run $0.94\text{--}1.24\times$ across 5 factors. This is the effect the benchmark was built to detect, and it is absent. The null rests on a derivatives-based flow proxy and cannot separate a genuinely stable capacity from an uninformative proxy.

6. Did validating on familiar strategies suffice?**[GENERATION METHOD]**

No, and this is the paper's methodological result. Applying the finished benchmark to 156 machine-written strategies *before* publishing it exposed two defects that 5 standard factors had not. **Repairing them moved every capacity figure downward.** The machine corpus spans $30.9\times$ against the factors' $9.8\times$ — $3.2\times$ wider — and both defects lived in tails the validation set never reached.

One programme, three questions. Each paper answers the question the previous one raises, on one shared universe, one shared cost model and one shared backtest engine.

- | | | |
|----------|---------------------|--|
| 1 | AlphaAudit | Can an AI produce a trading strategy that is real, rather than a plausible artefact of luck and leakage? |
| 2 | RegimeStress | If a strategy looks real, when does it break? |
| 3 | FlowState | If a strategy survives, how much capital can it absorb? |

Code → validity → robustness → scale. The programme's overarching question is whether making the AI better makes the *research* better — and the three papers answer it only to the extent their experiments support.

Contents

Key findings	1
1 The question	4
1.1 What this paper does and does not claim	5
2 What was available, and why it was insufficient	5
3 The benchmark	6
3.1 Universe, window, and what is held out	6
3.2 The factor zoo, and the two families missing from it	6
3.3 Deployment capacity, defined	7
3.4 Reproducibility, and how every number here was produced	7
4 Results	7
4.1 Deployment capacity, and what sets it	7
4.2 Alpha decay	9
4.3 Flow-conditional capacity	10
4.4 Why daily OHLCV cannot price transient impact	10
4.5 The validation gap	12
5 Does any of this depend on who wrote the strategies?	13
6 Design flaws, mistakes, and things that did not work	15
6.1 Results that failed	15
7 Threats to validity	16
8 What this tells us about AI-driven research	16
9 The FlowState benchmark	17
10 The companion benchmarks	17
11 Conclusion	18
Where this leaves us	18

Contributions

1. **A measured negative result on identification.** Daily OHLCV cannot pin down a transient-impact coefficient. We demonstrate this four ways—weighting-dependence of the sign, a resolution argument, an unstable exponent, and an extrapolation gap—each with a detectability bound so that absence of evidence is separated from evidence of absence, and each against a control measure that does remain stable.
2. **A capacity definition that survives the negative result.** Constraint-based deployment capacity requires no impact model, is what a pre-trade risk system actually enforces, and is a strictly smaller claim than the one the data cannot support.
3. **Deployment capacities for standard factors on Indian equities**, which we believe are not openly published, together with the mechanism that sets them.
4. **A null on flow-conditional capacity**, reported with the limitation of its proxy in the same breath as the finding.
5. **A methodological result on benchmark validation:** a benchmark validated on a narrow, familiar strategy family failed to expose defects that a broader machine-generated population exposed immediately. We adopt the resulting ordering as standing process.
6. **A frozen, reproducible artifact** in which no published figure is transcribed. Every number in this paper resolves through a macro generated from an artifact, and a check fails the build if a bare numeral appears in a claim position.

1 The question

Paper 1 asked whether a machine-written strategy is real, and for most of a 14.6%-executable corpus the answer was no. Paper 2 took the survivors and asked when they break, finding that fragility is a property of the strategy rather than of the model that wrote it, and that it cannot be predicted from strategy characteristics. Both questions are ones a researcher asks. **This paper asks the first question an allocator asks**, and it is the one neither predecessor could answer: *how much money can this take?*

A strategy that is real and robust at one crore may not exist at a hundred. Size is not a footnote on a backtest — it is a separate property, it is not implied by the Sharpe ratio, and a strategy can pass every test in papers 1 and 2 and still be undeployable.

Every strategy has a size at which it stops working. Run a hundred crore through a mid-cap reversal signal and the signal is intact; run a thousand and the buying required to establish the position is itself the reason the position is unprofitable. The number at which that transition happens is the single most consequential fact about a strategy that a backtest does not report, and it is routinely either omitted or asserted.

The standard construction is a capacity curve: net return as a function of deployed AUM, declining as the strategy's own trading pushes prices against it, with capacity read off where the curve crosses zero. Every point on that curve is a claim about market impact—how far a given order moves the price—and impact is not observable in a daily bar. It is therefore modelled, usually with the square-root law (Almgren et al., 2005), whose functional form is standard and whose coefficient is calibrated on proprietary records of institutional orders in which the order and the price path it caused are separately visible.

We have no such record, and neither does anyone working from public Indian data. The question this paper opens with is whether the coefficient can be recovered anyway.

It cannot, and Section 4.4 is the measurement rather than the assertion. The obstruction is not noise. It is that a daily bar aggregates every participant's trading and cannot isolate one order, and—more decisively—that temporary impact decays over minutes to hours, so the finest resolution a daily bar offers is already coarser than the phenomenon being estimated.

1.1 What this paper does and does not claim

1. **It measures whether daily bars can price market impact**, four ways, each against a control computed on the same bars. They cannot. §4.4.
2. **It redefines capacity as a constraint** that assumes no impact coefficient, and reports deployable size for standard factors on Indian equities. §4.1.
3. **It fails to find flow-conditional capacity**, the effect it was built to detect, and states the proxy that limits the null in the same breath. §4.3.
4. **It tests whether any of this depends on the generator**, using the same three frontier models papers 1 and 2 used. §5.

What this paper is not. It is not a capacity curve. We do not report the AUM at which a strategy's edge reaches zero, because that quantity is not identified from the data available to us, and reporting it would require an impact coefficient we would have had to invent. A great many published backtests price market impact from daily bars; this paper is in part a measurement of what that assumption is worth.

It is not a claim that impact does not exist, or that it cannot be measured. It is a claim about what *daily bars* can identify. Intraday data would change the answer, and we report in Section 7 what we could and could not obtain.

It is not a study of Indian market microstructure. It is a benchmark: a frozen, versioned instrument that returns the same verdict on the same strategy tomorrow, so that different strategy generators can be compared against a fixed standard.

2 What was available, and why it was insufficient

Impact models (Almgren et al., 2005) give us the functional form and are not our contribution. What they do not give us is the coefficient, which in every published calibration comes from proprietary records in which one institution's order and the price path it caused are separately visible. The question nobody appears to have measured is what happens when that record does not exist, which is the situation of everyone working from public Indian data. §4.4 is that measurement.

Liquidity measures (Amihud, 2002) are computable from daily bars and are stable on ours. That is exactly why Amihud illiquidity is reported here as a *control* rather than as a result: it establishes that the data is adequate, so that the failure of the impact parameter cannot be attributed to noisy bars.

Trading-cost taxonomies (Novy-Marx & Velikov, 2016) establish that costs reorder the factor zoo and that many published anomalies do not survive them. They work on US data with US microstructure, and they price impact from an assumed coefficient — the assumption this paper measures.

Backtest methodology (López de Prado, 2018) supplies the discipline papers 1 and 2 are built on, and its concern is whether a result is real. Capacity is a different axis: a result can be entirely real and still not be available at size. Nothing in that literature tells an allocator how much money a signal absorbs.

The gap FlowState fills: there was no released benchmark reporting deployment capacity for Indian equities from observable liquidity, and no published measurement of whether daily bars can identify the impact coefficient that the standard capacity construction requires.

3 The benchmark

3.1 Universe, window, and what is held out

The universe is rules-based rather than index-based. At each quarterly rebalance it is the top names by trailing median traded value, computed only from sessions strictly prior to that date. Nothing consults a published membership list, which matters because a universe assembled from today's index constituents and applied backwards is survivorship-biased by construction: firms that were dropped after performing badly simply never appear. Under a rules-based definition a firm that deteriorates stops qualifying at the next rebalance and its bad period stays in the record.

Over 2020–2024 this yields 184 symbols and 122,464 investable symbol-days. A holdout year is reserved and **was never opened by any part of this project**.

3.2 The factor zoo, and the two families missing from it

Five signals, all constructible from price and volume alone: twelve-month momentum skipping the most recent month (Jegadeesh & Titman, 1993), one-month reversal, low realised volatility, Amihud illiquidity (Amihud, 2002), and a traded-value size proxy. Books are equal-weighted quintile long-short portfolios, formed every session, rupee-neutral by construction so that capacity scales linearly in deployed AUM.

Every signal is point-in-time by construction rather than by convention: each uses a trailing window ending strictly before the session it labels, enforced by shifting before rolling. A test truncates the price panel and re-derives every score, requiring bit-identical values for every session at or before the cut. A factor that peeks at its own future is indistinguishable from a good factor until somebody checks.

Value and quality are absent, and their absence is a scope limitation rather than an oversight. Both require company fundamentals—book value, earnings, accruals, return on equity—which this repository does not hold and could not acquire under a market-data budget fixed at zero. Any comparison against published factor results must be read knowing that the two most fundamentals-dependent families are missing. For the same reason the fifth factor is named `liquidity_size_proxy` rather than `size`: it is built from trailing traded value, not market capitalisation, which needs a share count we do not have. The earlier name invited a comparison against published size-factor results that would not have been like-for-like.

3.3 Deployment capacity, defined

For a book rebalancing from weights w to w' , the traded fraction in name i is $|\Delta w_i|$. At deployed AUM A the rupee trade is $A|\Delta w_i|$, and the participation constraint permits at most κ of that session's traded value $V_i(t)$. Hence

$$A_{\max}(t) = \min_{i: |\Delta w_i| > \varepsilon} \frac{\kappa \cdot V_i(t)}{|\Delta w_i(t)|}, \quad A_{\max} = \min_t A_{\max}(t).$$

Three details in that expression are load-bearing.

$V_i(t)$ is a trailing average, not the session's own volume. Sizing an order against the volume that the order itself helped create is circular, and the circularity flatters capacity.

The outer minimum runs over sessions, not a median. A strategy must be executable on every rebalance it makes, not on a typical one. This also makes the session that builds the opening position binding for a near-static strategy, so initial portfolio construction is priced like any other trade rather than treated as free. Section 4.5 describes what a median concealed before we corrected it.

ε excludes weight changes too small to be trades. A strategy that recomputes an unchanged weight produces differences of order 10^{-18} , and dividing a participation limit by that residue produces a capacity of 10^{23} crore. We set $\varepsilon = 1.0 \times 10^{-9}$ of the book.

3.4 Reproducibility, and how every number here was produced

No number in this paper was typed. Each figure is computed by a generator from a frozen artifact, emitted as a macro, and referenced by name; a companion check parses the paper, strips structural markup, and fails the build if any bare numeral survives in a claim position. Numeric literals that are genuinely not results—a holding horizon, the power of a statistical test—appear on a short allowlist, each with a written reason. A number that is genuinely a measurement cannot be added to that list honestly.

The gate is tested by breaking it. A test takes the committed paper, substitutes one macro for the value it renders to—exactly as a careless edit would—and requires the check to start failing. A gate nobody has seen fail is indistinguishable from a gate that cannot fail, so the control runs on every commit rather than existing as a one-time manual act.

Each paper in this programme owns a macro namespace, enforced by a further test, so that a figure generated for one paper can never silently satisfy a reference in another. Two papers would then share a number without sharing an artifact, which is the precise failure the apparatus exists to prevent. The global seed is fixed by the project charter; every script writes a run manifest recording git SHA, configuration hash, package versions and input file hashes.

4 Results

4.1 Deployment capacity, and what sets it

Takeaway. The weakest position determines the capacity of the entire portfolio. A book is not deployable at its average liquidity — it is deployable at the liquidity of the single worst trade it must place, and the gap between those two readings runs from $4.3\times$ to $26.2\times$.

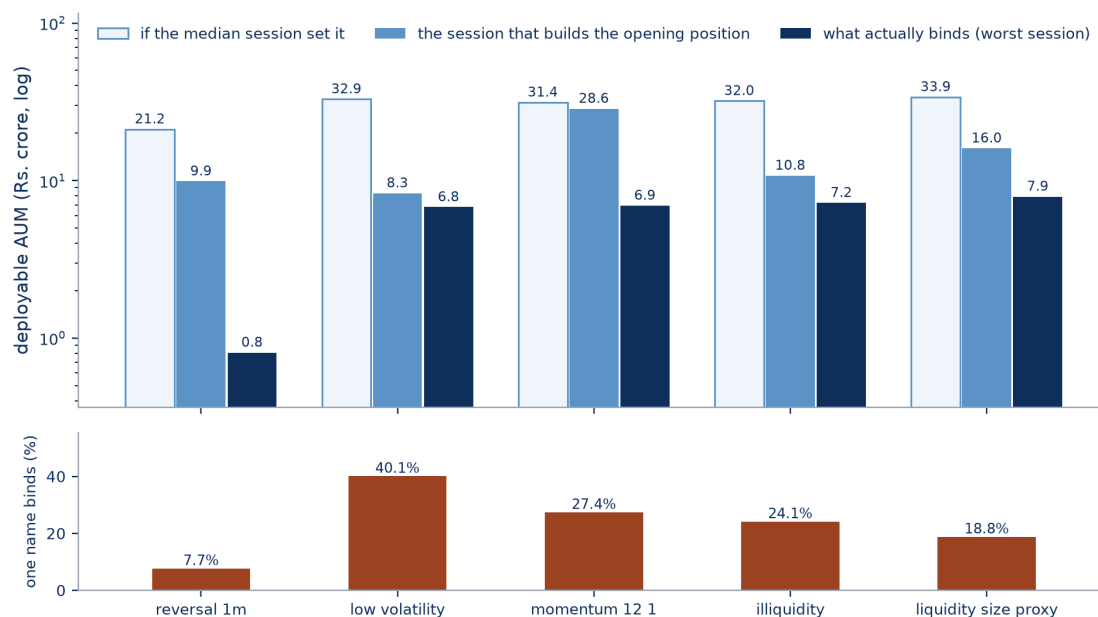


Figure 1. Deployment capacity in crore of rupees at a 1.0% participation limit, across 5 factors. *Upper panel, log scale:* what capacity would be if a typical session set it (pale), the session that builds the opening position (mid), and what actually binds (dark). The distance between pale and dark is what a typical-session summary would have concealed. *Lower panel:* the share of sessions on which the tightest position is more than twice as tight as the next — that is, on which a single name is the constraint.

Deployment capacity is **Rs. 0.81–7.91 crore**, roughly one to ten million US dollars. **The least liquid position the strategy holds decides it, not the average liquidity of its book.**

The mechanism, because the number is otherwise easy to misread. These are equal-weighted books. Every name in a leg carries the same weight and must therefore be traded in the same rupee size. A strategy holding a name that turns over a tenth of what another turns over trades both identically, and the participation constraint binds on the thinner one first. Capacity is set there, for the whole book.

The consequence is counter-intuitive and worth stating plainly: **a strategy holding a hundred large-capitalisation names can still be capacity-constrained by one of them.** Diversification across many liquid names does not raise capacity if the weighting scheme forces the thinnest name to be traded at the same size as the deepest. The *one name* column reports how often this is literally a single position.

This is not an artefact of our definition—it is the constraint a desk actually faces—but it is a property of *equal weighting* as much as of the market. The standard remedies, liquidity-tilted weights or a liquidity screen at entry, both change the portfolio and therefore change the returns that the capacity is a capacity *for*. We measure the strategies as specified rather than silently repairing them, and note that a capacity figure is only meaningful when paired with the weighting scheme that produced it.

Dropping the single tightest position each session raises capacity by well under an order of magnitude. That robustness check is what establishes these figures describe the strategy rather than one holding; had relaxation bought tenfold, the headline would have been a statement about one name.

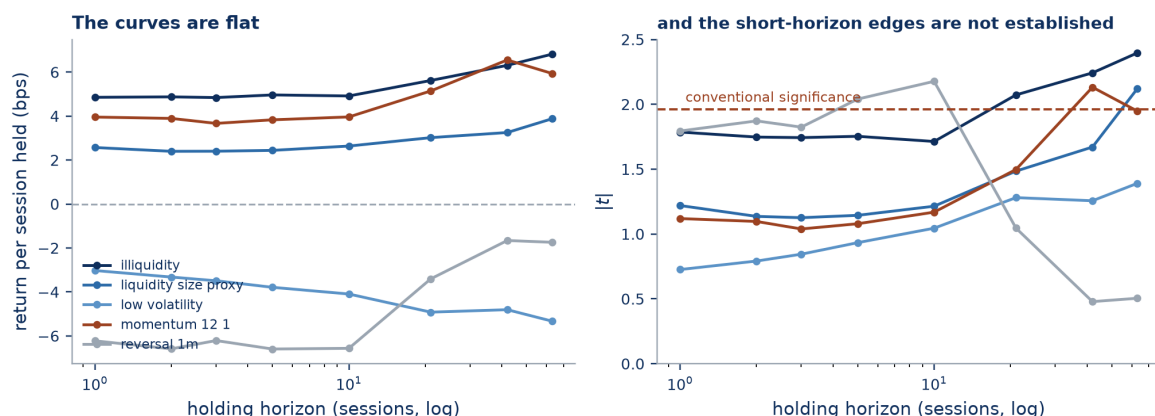


Figure 2. Long-short return per session held, gross of costs, across 1,233 formation dates. Books are formed every session and held for h sessions after a one-session execution gap, so consecutive observations overlap; t -statistics divide by the number of independent holding periods rather than the raw count, which is deliberately conservative. *Left:* a decaying edge would fall from left to right. None does. *Right:* the same factors against the conventional threshold, which no factor crosses at any horizon short enough for the question to be about decay.

4.2 Alpha decay

0 of 5 factors have a measurable half-life. Every curve is flat to gently rising out to a quarter.

We report this as a null about our measurement rather than as a finding about Indian markets, because the precondition fails. The largest $|t|$ at the shortest horizon is 1.79, and the largest at any horizon in the table is 2.40. **The edges whose decay is being measured are not themselves established**, and a decay half-life is only a meaningful quantity for a signal that had an edge to lose. Reporting one for a factor that never made money would be a category error dressed as a measurement.

Two factors carry the wrong sign for their names, which is itself informative about the window. `reversal_1m` loses money, so at a one-month horizon this period shows *continuation* rather than reversal. `low_volatility` loses, so high-beta names outperformed—consistent with a window containing the COVID drawdown and the recovery that followed it, in which the names that fell hardest rebounded hardest.

A research question retired rather than deferred. This benchmark was specified to ask whether alpha decays faster in India than in developed markets and whether retail participation explains it. The first clause is answered: no decay was observed. The second therefore loses its antecedent—there is nothing for retail participation to explain—and is retired. Delivery-share data that would have supported a retail analysis is freely available and was deliberately not collected, on the principle that the availability of data is not a reason to ask a question.

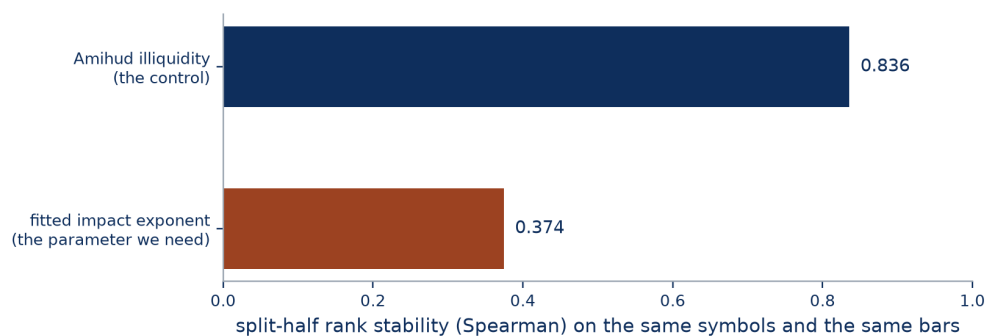


Figure 3. The control and the parameter, on the same daily bars, the same symbols and the same sample split ($n = 166$). Rank stability across the split.

4.3 Flow-conditional capacity

Outflow-to-inflow capacity ratios lie between 0.94 and 1.24 across 5 factors, median 0.96, on 76–234 sessions per cell. **The hypothesis is not supported.**

The novel contribution this benchmark was built for was to ask whether deployable size collapses when foreign participation reverses. India is a plausible place to look: foreign flows are large, volatile, and widely believed to drive liquidity. A tenfold swing in capacity between inflow and outflow regimes would be a useful and previously unquantified fact.

It is not there. The ratios are within a quarter of unity and some point the wrong way.

This null rests on a proxy, and we cannot separate two explanations. No free source serves historical daily foreign and domestic institutional *cash-market* net flow. The exchange publishes the series but its endpoint returns the current session regardless of the date requested, which we verified against three parameter spellings; the alternative sources we tried returned error pages or refused the request. The series we use instead is foreign-institutional activity in *equity derivatives*, and it differs from the intended series in three ways that are not small. It is denominated in contracts rather than rupees, over a window in which regulatory lot sizes changed. It describes derivatives rather than cash. And foreign derivatives activity is substantially hedging against cash positions, so its sign need not track the direction of cash-market demand at all.

Whether capacity is genuinely stable across flow states, or the proxy is simply uninformative about cash liquidity, is not separable with this data. We state this beside the result rather than in a limitations section, because it governs how the null may be read.

4.4 Why daily OHLCV cannot price transient impact

Takeaway. This single comparison carries the negative result. A quantity computed from exactly the same data holds its ordering at 0.836 while the parameter a capacity study needs reaches 0.374. **If the data were simply too noisy, both would fail. The failure is in the model.**

A price move caused by an order is *transient*: it is the cost of demanding liquidity and it is repaid once the demand stops. A move caused by information is permanent. A negative reversal coefficient is therefore the signature of impact, and its sign is the quantity on which any daily-bar impact estimate depends.

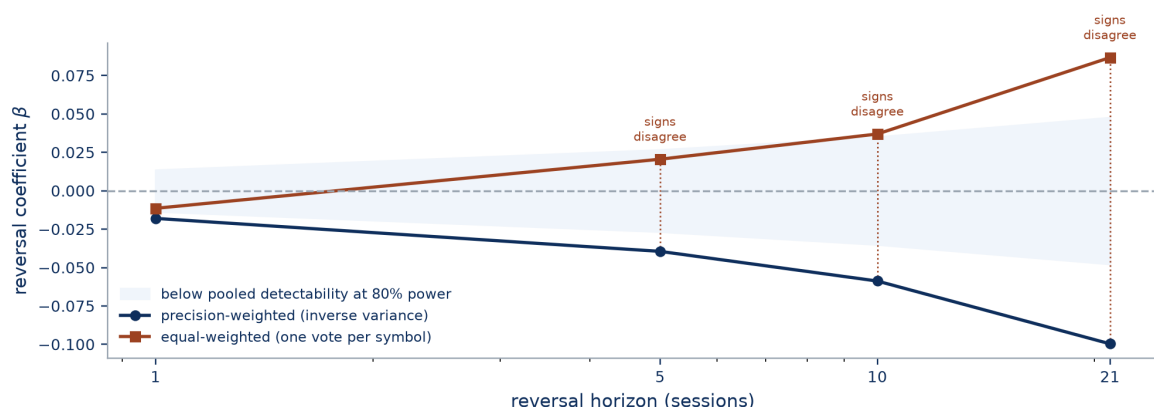


Figure 4. Reversal regressions (Campbell et al., 1993) of the following h -session return on today's, restricted to each symbol's heaviest-volume decile, $n = 177$ symbols. Precision weighting is inverse-variance pooling; equal weighting counts each symbol once. Both are standard and neither is wrong. The shaded band is the effect the pooled estimator could have detected at 80% power, so the two lines are separated by more than the data's own resolution — the disagreement is not a power problem, and **the sign of the coefficient is decided by the analyst rather than by the market.**

The sign depends on the weighting scheme. Precision weighting and equal weighting are both standard, neither is wrong, and they **disagree on sign at 3 of 4 horizons**. A quantity whose sign is determined by an analyst's choice among defensible estimators is not identified by the data.

The disagreement is not a puzzle to be resolved by picking the better estimator; it is the result. Inverse-variance pooling is correct when symbols estimate a common parameter and differ only in noise. Equal weighting is correct when the parameter itself varies across symbols and the quantity of interest is its population average. **Nothing in the data tells us which world we are in**, and the two answers have opposite signs. Choosing between them requires prior knowledge of exactly the microstructure the exercise was supposed to measure.

Three further measurements agree, and one control locates the failure:

- **Resolution.** Temporary impact decays over minutes to hours; the finest horizon a daily bar permits is one session. The pooled coefficient also grows monotonically more negative as the horizon lengthens rather than decaying, which is the signature of slow mean reversion, not of a liquidity concession being repaid.
- **Exponent.** Fitting $|r|$ against relative traded value gives a median exponent of 0.689 against the square-root law's 0.5, spanning 0.521–0.859 across 179 symbols and explaining a median 0.126 of variance. Across a sample split the per-symbol exponent correlates 0.374 with itself ($n = 166$). A parameter that does not survive its own sample split is not a parameter.
- **Extrapolation.** A capacity study asks about 1.0% participation. Observed relative traded value has a median of 0.821–1.91 orders of magnitude away—and only 2 of 208,042 symbol-days fall inside the region of interest. Fitting here and applying there is not interpolation.
- **The control.** Amihud illiquidity, computed on the same panel, correlates 0.836 across the same split ($n = 166$, $p = 1.7 \times 10^{-44}$). The data supports a stable liquidity *measure* while refusing to support an impact *model parameter*. **The failure is in the model, not in the data**—which is the whole reason the control is reported.

Two corrections to our own earlier reporting of this result. An earlier version reported that *no* reversal was detected, citing the proportion of symbols with a negative coefficient as evidence. That proportion was noise. A single symbol's daily test can detect only an effect of 0.368 or larger at the five-session horizon—an order of magnitude above the pooled coefficient of -0.0395 that the same data supports—so the proportion was always going to land near half regardless of the truth. We had reported an underpowered statistic as a measurement.

Computing the detectability bound then showed that the pooled coefficient *is* significantly negative, contradicting our own earlier claim outright. The identification argument above replaced it and is stronger. We record the sequence because the null we first reported was not the null that survived, and a reader is entitled to know that the argument moved.

4.5 The validation gap

The benchmark was validated on 5 standard factor strategies spanning $9.8\times$ in capacity, then applied to 156 machine-generated strategies spanning $30.9\times$ — $3.2\times$ wider. **Two defects lived in that gap, and the corpus found both immediately.**

Before freezing the benchmark we ran it across a corpus of strategies generated by a local language model and previously evaluated by two companion benchmarks in the same programme. The intent was to produce a reference corpus in which every strategy carries an audit verdict, a fragility score and a deployment capacity; 125 strategies do. What it produced first was two bugs.

Floating-point residue counted as trading. A strategy holding a hundred names at equal weight and never rebalancing produces weight changes of order 10^{-18} from recomputing an unchanged number. Dividing a participation limit by that residue reported its capacity as 10^{23} crore. The remedy is the tolerance ε defined earlier.

The summary statistic discarded the binding constraint. Capacity was reported as the *median* session. For a near-static book that is close to meaningless: almost every session has near-zero rebalancing turnover, so the one tightly constrained session that builds the position was drowned by a thousand that barely trade. The remedy is the minimum over sessions.

A subtlety worth recording, because we diagnosed it wrongly at first. We initially believed the formula never counted the acquisition of the opening position. It did—a strategy's first session enters at full weight from zero. **The defect was that the summary statistic discarded it**, which is a different error and took a direct check to see. Using the median would overstate the typical corpus strategy by $10.6\times$, and entry is the binding session for only 32 of 156 strategies (20.5%)—so the median was concealing considerably more than the entry constraint alone.

Neither defect was reachable from the validation set. All five factor strategies rebalance substantially every session, so none ever produced a near-zero traded fraction, and for all five the median and the minimum differ by a factor small enough to look like a modelling choice rather than a bug.

The corrections moved every published number down. Capacity figures fell by between a factor of four and a factor of twenty-six. We record the direction because it bears on whether this counts as tuning: adapting a benchmark to flatter its subject would not systematically

reduce the results it reports. Repairing an objective omission in a definition is not the same act as adjusting a threshold until a result improves, and the sign of the change is the cheapest available evidence of which one happened.

We state the obvious objection rather than leave it to the reader to raise. *Both defects were found by data the benchmark was about to be applied to, and repaired before it was frozen*—the corrections are, in the literal sense, post-hoc. Our argument that they are repairs and not tuning rests on three things: each defect is demonstrable without reference to any result, in one case as a dimensional absurdity and in the other as a statistic that provably omits a constraint the definition already contained; the ordering was fixed in advance, so the corpus run was scheduled before the freeze rather than reached for after a disappointing number; and every correction moved every published figure downward. None of that is proof, and a reader is entitled to weigh it themselves. We report it here, in the result, rather than in a list of caveats, because the alternative—discovering the defects after publication, or not at all—is the outcome this ordering exists to prevent, and a paper that buried the episode would be recommending a process it declined to display.

The general form: **a benchmark’s soundness cannot be established on the population that motivated it.** Defects concentrate in the tails of the population a benchmark is applied to, and a validation set assembled from familiar, well-behaved strategies has no tails.

We have adopted the resulting ordering as standing process for every benchmark in this programme: **build, validate on standard strategies, apply to the reference corpus, freeze, write up.** The corpus run precedes the freeze and the paper. Had the original ordering held, this paper would have shipped a capacity table containing 10^{23} crore.

5 Does any of this depend on who wrote the strategies?

Every capacity figure above was measured on strategies written by one local seven-billion-parameter model. The objection a referee reaches for first is that this measures that model rather than machine-written strategies in general, and it is a fair objection: nothing in the preceding sections distinguishes the two.

We therefore issued the identical frozen task specification to 3 frontier models through their consumer interfaces, 4 independent requests of five strategies each, and passed all 60 through this benchmark with no parameter altered and no line of measurement code added. The design, the hypotheses and the predicted directions were pre-registered before any strategy was generated. Full results, including the measurements that belong to the other two benchmarks, are in the study’s own report; what follows is the part that bears on capacity.

The pre-registered prediction failed on one arm of three. We predicted that median deployment capacity would fall within a factor of two of the local corpus for every generator, on the reasoning that capacity is set by the traded value of Indian large caps and by the participation limit, neither of which knows who wrote the code. Two arms are consistent with that: GPT at $1.15\times$ and Claude at $1.97\times$. Claude clears the stated bound by a margin far smaller than the sampling error of a twenty-strategy median, and we report it as not clearly failing rather than as passing. Gemini Pro is $0.31\times$ — outside the bound, and in the opposite direction from the one the bound was written to catch.

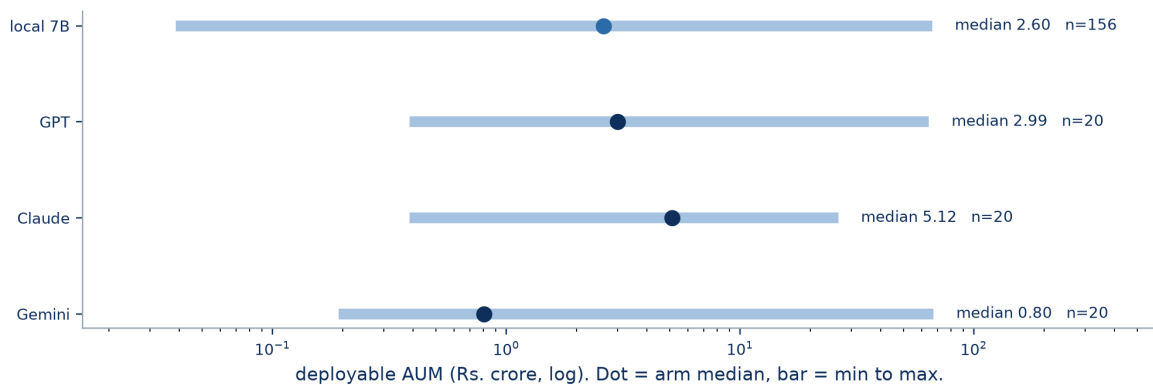


Figure 5. Constraint-based deployment capacity by generator, log scale. The dot is the arm median and the bar is its full range. The local corpus is 156 strategies; each frontier arm is twenty. **The medians move — one of them outside the pre-registered bound — while every arm spans one to two orders of magnitude internally.** The second fact is the more important one: the spread within any single generator is wider than the gap between generators.

The flow-conditional null holds too. Section 4.3 reported that deployable capital does not collapse when foreign investors withdraw — a prediction we made and then falsified on five factor strategies. Sixty machine-written strategies agree. The ratio of outflow capacity to inflow capacity has a median of 0.952 on the GPT arm, 0.868 on Claude and 0.905 on Gemini, against 0.96 for the factors. No arm’s median falls below 0.868. Per-strategy ranges are 0.770–1.041 on the GPT arm, 0.700–1.116 on Claude and 0.615–1.414 on Gemini, so not one of the 60 strategies loses even half its deployable size when flows reverse. A population that spans two orders of magnitude in capacity therefore reproduces a null established on one that spans less than ten times, which is the strongest evidence this benchmark has that the flow-conditional result is a property of Indian market liquidity rather than of the five strategies it was first measured on.

What we do not claim. We have not investigated why. An account of Gemini’s narrower books would be constructed after seeing the result it explains, and would be exploratory in the precise sense this laboratory exists to police. We state the measurement and stop.

What it means for this paper. Capacity is *not* generator-invariant, and any reading of Section 4.1 as a statement about machine-written strategies in general is unsupported by this evidence. The figures in this paper describe the strategies this paper measured. That is a narrower claim than we expected to be making, and it is the claim the data supports.

What survives, and is strengthened. The *mechanism* is unchanged across all four populations: capacity is set by the least liquid name a strategy insists on trading, the span within each arm is one to two orders of magnitude, and the frontier arms span $349.7\times$ at the widest against $30.9\times$ locally. The validation gap of Section 4.5 therefore recurs with more force than it was first reported with: a benchmark validated on five factor strategies spanning under ten times was, on this evidence, being validated on a population narrower than any generator produces.

6 Design flaws, mistakes, and things that did not work

A benchmark that reports only its successes is not measuring anything. Everything below is a defect in our own apparatus, a capability we withdrew, or a result that failed.

Table 1. Defects, withdrawals and nulls, with what each cost.

Defect, withdrawal or null	What it cost	Where
Floating-point residue counted as trading	A strategy recomputing an unchanged weight produces differences of order 10^{-18} ; dividing a participation limit by that residue produced absurd capacities. Found only when the machine corpus was run.	§4.5
The headline statistic discarded the binding constraint	Reporting a median session rather than the worst one overstated capacity by up to $26.2\times$. A strategy must be executable on <i>every</i> re-balance.	§4.5
The price panel was truncated	A loader discarded the warm-up rows a lookback strategy needs on its first session. Recorded as a correction rather than treated as invalidating the published figures.	corrections log
The ML impact model was withdrawn	Originally specified here. It would have been trained to predict a quantity the data cannot identify, and benchmarked against an equally unidentified baseline — a comparison between two artefacts. Deleted rather than shipped.	§4.4
Value and quality factors are absent	Both need company fundamentals this repository does not hold and could not buy at a zero budget. The zoo is price-and-volume only, and every comparison against published factor results must be read knowing the two most fundamentals-dependent families are missing.	§3.2
The impact coefficient stays uncalibrated, permanently	Not a to-do. §4.4 is the measurement showing it cannot be calibrated from this data, so the cost model carries it as a stated engineering assumption forever.	§4.4

6.1 Results that failed

- **No factor has a measurable decay half-life:** 0 of 5. None is statistically established even at the shortest horizon (largest $|t| = 1.79$), so the precondition for a decay study fails. This was one of the project’s three research questions.
- **Capacity does not collapse under foreign outflows,** the effect this benchmark was designed to detect.
- **Daily bars cannot price transient impact,** which was the original form of the headline question.
- **Validating on standard factors was insufficient,** which we found only by doing the thing that exposed it.

Three of this paper’s four original research questions returned nulls, and the fourth — capacity — survives only because it was narrowed to a claim the data supports. **We report the narrowing in the results rather than presenting the smaller claim as though it had always been the plan.**

7 Threats to validity

1. **The flow proxy.** The flow-conditional null rests on derivatives activity rather than cash-market flow. This is the threat we would attack first, and it is stated in the abstract and in the result rather than only here.
2. **Capacity is an extreme-value statistic.** The headline is a minimum over roughly twelve hundred sessions and is therefore sensitive to any one of them. We report the fifth percentile and a drop-the-tightest-name variant alongside it, but a minimum is a minimum and should be read as one.
3. **The factor zoo is incomplete.** Value and quality are absent for want of fundamentals, which weakens any comparison against published factor results and means the zoo over-represents liquidity-driven signals.
4. **Decay is measured on edges that are not established.** With no factor significant at the shortest horizon, the decay curves describe the decay of quantities that may be zero. A flat curve through zero is not evidence of persistence.
5. **Returns are gross of costs.** Indian transaction costs are large enough to reverse the sign of a small edge, and every decay figure here would fall under them.
6. **Equal weighting drives the capacity result.** The finding that the least liquid position binds is a joint property of the market and of the weighting scheme. A different scheme would give different capacities for the same signals.
7. **One market, one window.** 2020–2024 in Indian equities contains one pandemic drawdown and one recovery. Every negative result here may be a property of that window.
8. **Intraday data would change the identification result.** We established that no free source reaches this window at intraday resolution, and that commercially available sources at a retail price point reach only part of it. With minute bars the extrapolation gap closes and the transience test can be run at the scale impact actually operates. Our claim is about daily bars, not about the phenomenon.

8 What this tells us about AI-driven research

Three claims, at three scopes, kept separate — the same discipline papers 1 and 2 apply, because the three papers must be readable as one argument.

Generator-independent [GENERATOR-INDEPENDENT] The *mechanism* that sets capacity, the flow-conditional null, and the identification failure are all properties of the market rather than of the code put through it. The identification result does not involve a strategy at all. These are the claims that license the benchmark being used by other people.

Generator-specific [LOCAL 7B] The capacity *levels* are not generator-invariant. §5 falsified our own pre-registered prediction on one arm of three, and the honest reading is that the figures in §4.1 describe the strategies this paper measured rather than machine-written strategies in general. That is narrower than we expected to be claiming.

Method-independent [GENERATION METHOD] The validation gap is not about language models at all. A benchmark validated on a narrow, familiar population missed defects that a broad one exposed immediately, and the generality of that has nothing to do with what wrote the strategies — only with how much of the input space the validation set covered.

Paper 1 found that a better generator writes better code without producing better evidence. Paper 2 found that it does not write more robust strategies either. This paper completes the pattern from the other direction: **the constraint that decides how much money a strategy can absorb is set by market liquidity, and no amount of generator capability moves it.** Across three papers, capability moved the code and did not move the evidence.

9 The FlowState benchmark

Frozen and released at github.com/Ashwin-aadi/Trivijaya-Quant-Research:

- **The liquidity universe.** 184 symbols and 122,464 investable symbol-days over 2020–2024, rules-based and survivorship-free, with a holdout year that was never opened.
- **The capacity definition,** with its trailing-volume convention, its minimum over sessions, and the tolerance ε that the machine corpus forced us to add.
- **The factor zoo,** price-and-volume only, with the two families we could not build named rather than quietly omitted.
- **The identification suite** — four diagnostics, each with a detectability bound, and the Amihud control that locates the failure in the model rather than the data. This is shipped as an artefact because it is the result other researchers are most likely to want to contest.
- **The reference corpus.** 125 strategies carrying an audit verdict, a fragility score and a deployment capacity, measured on one universe under one cost model.
- **Every number,** regenerable from `data/raw/` by documented commands. If this paper and the repository disagree, the repository is right.

10 The companion benchmarks

This benchmark is one of three built on a shared point-in-time universe, a shared Indian transaction-cost model and a shared backtest engine, released from a single repository. A strategy can therefore carry an audit verdict, a fragility score and a deployment capacity measured on the same data under the same assumptions, which is what makes the cross-benchmark statements in this paper possible.

- **AlphaAudit** — whether a generated strategy is real, or an artifact of leakage and multiple testing. It supplies the strategy corpus the other two measure.
[papers/pdf/1-alphaaudit.pdf](#) · [benchmarks/alphaaudit](#)

- **RegimeStress** — when a surviving strategy breaks, measured across counterfactual regime paths. It supplies the fragility scores and the duplicate criterion.
[papers/pdf/2-regimestress.pdf](#) · [benchmarks/regimestress](#)
- **Generator validation** — the study of Section 5: the identical frozen task specification issued to three frontier models and passed through all three benchmarks unchanged, with its pre-registration, its falsified hypotheses and its full results.
[benchmarks/generator_validation](#)

The three were designed to be read in order — is it real, when does it break, how much can it take — and each answers a question the previous one raises. Every artifact, correction log and evaluation protocol is in the repository: github.com/Ashwin-aadi/Trivijaya-Quant-Research.

11 Conclusion

FlowState reports what daily bars can support and declines what they cannot. Deployment capacity for standard price-and-volume factors on an Indian liquidity universe is Rs. 0.81–7.91 crore, set in every case by the least liquid position each strategy holds rather than by the average depth of its book. No factor's edge decays measurably within a quarter, and none is established at the shortest horizon. Deployable size does not collapse when foreign participation reverses, though that null is conditional on a proxy we would rather not have needed.

The result we expect to outlast the others is the one about instruments rather than about markets. Within the estimators and information daily OHLCV affords, a transient-impact coefficient is not identified—the obstruction is not statistical precision, but that two equally defensible estimators disagree about its sign and the data cannot arbitrate. A great many published backtests price market impact from daily data. This is a measurement of what that assumption is worth, and it comes with a control showing that the same data supports a stable liquidity measure perfectly well.

FlowState is intended as benchmark infrastructure rather than a completed scientific result. It is frozen, versioned, and reproducible from raw data by a documented command sequence, so that future strategy generators, future capacity definitions and future impact estimators can be evaluated against the same protocol and progress can be measured against a common standard. The corpus it now carries—125 strategies with an audit verdict, a fragility score and a deployment capacity apiece—exists to make the next comparison possible.

Where this leaves us

This paper closes the sequence the programme set out to build. A trading idea now passes through three questions in order — *is it real, when does it break, how much can it take* — and 125 machine-written strategies carry an answer to all three, on one universe, one cost model and one backtest engine.

The programme asked whether making the AI better makes the *research* better. Across three papers the answer is consistent and narrow. **Capability moved the code and did not move the evidence.** Frontier models took executability from 14.6% to complete and left out-of-sample performance, deflation and fragility where they were. No scaffolded generation method beat plain prompting at matched compute. And the constraint this paper measures — how much money a strategy can absorb — is a property of the market's liquidity, not of who wrote the strategy.

What we would not claim is that this settles anything about AI-driven research in general.

It is one market, one window, one apparatus, with the confounds each paper names. What the three papers do provide is the apparatus itself, frozen and public, so that the claims can be contested with measurements rather than opinions — including the several places where our own instruments turned out to be the thing that failed.

References

-
- Almgren, R., Thum, C., Hauptmann, E., & Li, H. (2005). Direct estimation of equity market impact. *Risk*, 18(7), 58–62.
- Amihud, Y. (2002). Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.
- Campbell, J., Grossman, S., & Wang, J. (1993). Trading volume and serial correlation in stock returns. *Quarterly Journal of Economics*, 108(4), 905–939.
- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Novy-Marx, R., & Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.